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Numerical Solution Of PDE- Constraints Optimization of Convection–Diffusion Equations

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Outline

- Introduction
- Formulation
- Optimization
- Discretization
- Solve system of linear equation
- Preconditioner
- Numerical Example

Big Picture

Problem Model (PDE)

Discretization

System of linear equation

- Direct Method (Projection methods):
- Iterative Method (Krylov methods)

Preconditioner

Fast Numerical Solution

The rate of convergence of a Krylov subspace method for a linear system ***Ax* = *b*** depends on the condition number of the matrix ***A*.**

Numerical Method for PDE

Finite difference schemes

- Main benefits
  - Simple to implement and fast
  - High-order is feasible
  - Local and explicit in time
  - Direction can be exploited - upwind
  - Extensive body of theory
- **Main problem**
  - Simple local approximation and geometric flexibility are not agreeable
  - Grid smoothness requirements

Numerical Method for PDE

Finite volume methods

- Main benefit
- Robust and fast due to locality
- Complex geometries
- Well suited for conservation laws
- Local and explicit in time
- Solid theoretical framework

Main problem

- Inability to achieve high-order on general grids due to extended stencils
- Grid smoothness requirements

Numerical Method for PDE

Finite element method

- High-order accuracy and complex geometries can be combined
- Extensive theoretical setting
- This yields the global equation

Main problems

- Implicit in time
- Not well suited for problems with direction

Hp-Finite element methods

Hp-FEM is a numerical method for solving partial differential equations based on piecewise-polynomial approximations that employs:

- elements of variable size (h)
- polynomial degree (p)

The performance of the hp-adaptivity in which error in energy norm is reduced by way of:

- mesh refinement (h)
- polynomial order extension (p)

Comparison Methods

	Complex geometries	High-order accuracy and hp-adaptivity	Explicit semi-discrete form	Conservation laws	Elliptic problems
FDM	×	✓	✓	✓	✓
FVM	✓	×	✓	✓	(✓)
FEM	✓	✓	×	(✓)	✓
DG-FEM	✓	✓	✓	✓	(✓)

What we need is a scheme that combines

- The local high-order/flexible element of FEM
- The local statement on the equation for FVM

These are exactly the components of the Discontinuous Galerkin Finite Element Method

Discontinuous Galerkin Method (DG-FEM)

For the application of numerical methods we want

- Accuracy at minimal effort
- Flexibility to solve different types of problems
- ‘Easy and quick’ prototyping
- Solid theoretical backing
- Good performance on advanced computers

Discontinuous Galerkin Method (DG-FEM)

The basics of DG-FEM

What did we learn from this?

- Stability is enforced through the flux choice.
- No restrictions on the local basis, e.g., it need not be polynomial, and is chosen to provide accuracy.
- The numerical solution is discontinuous between elements.
- Boundary conditions and interface conditions are imposed **weakly**.
- All operators are local.
- Due to the weak interface based coupling, there are no restrictions on element size and local approximation.

We know now

- It is the flux that gives stability
- It is the local basis that gives accuracy.

The scheme is very (VERY) flexible.
BUT - we have doubled the number of de

Optimization

The further information under publishing
and will be updated very soon